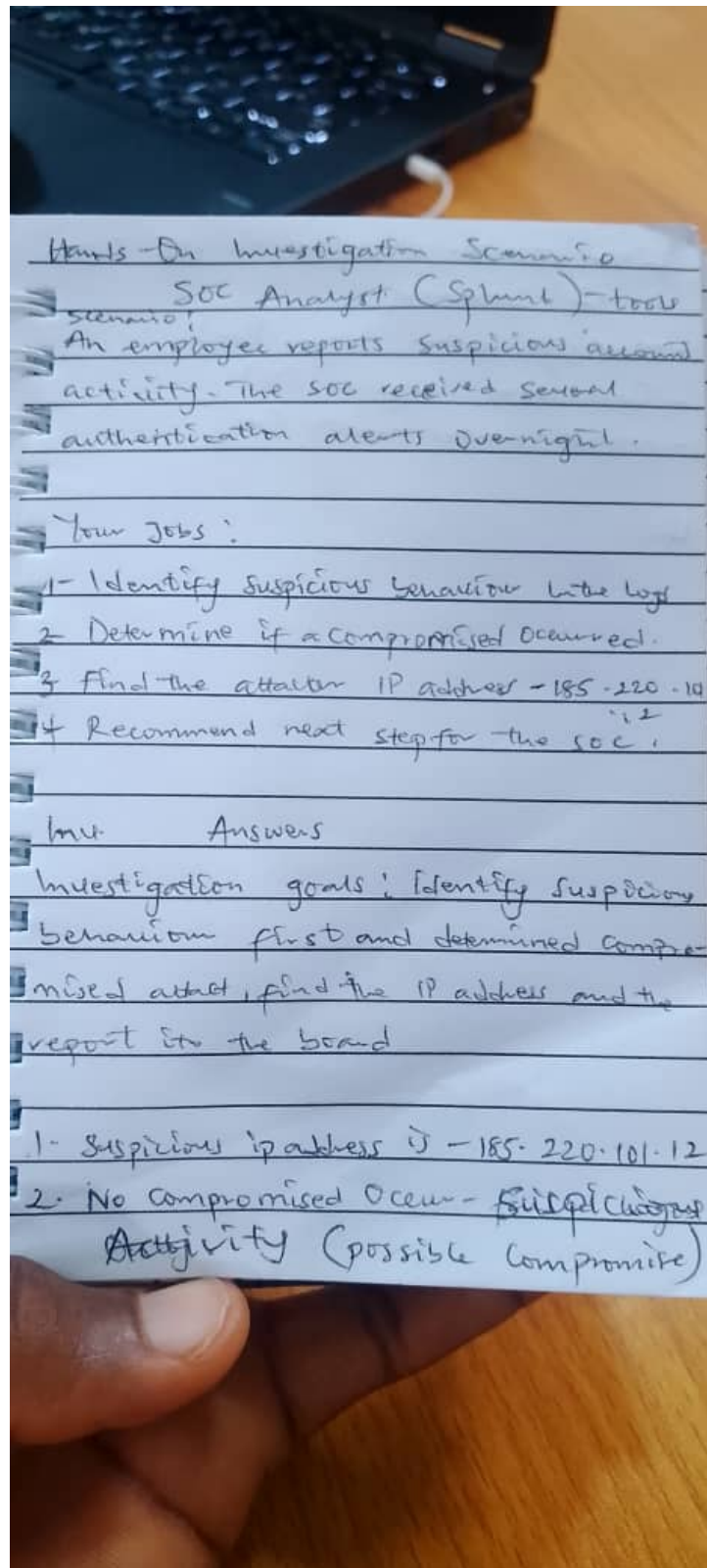


SOC Analyst Investigation Project Portfolio

Working Notebook – Investigation Scenario and Findings



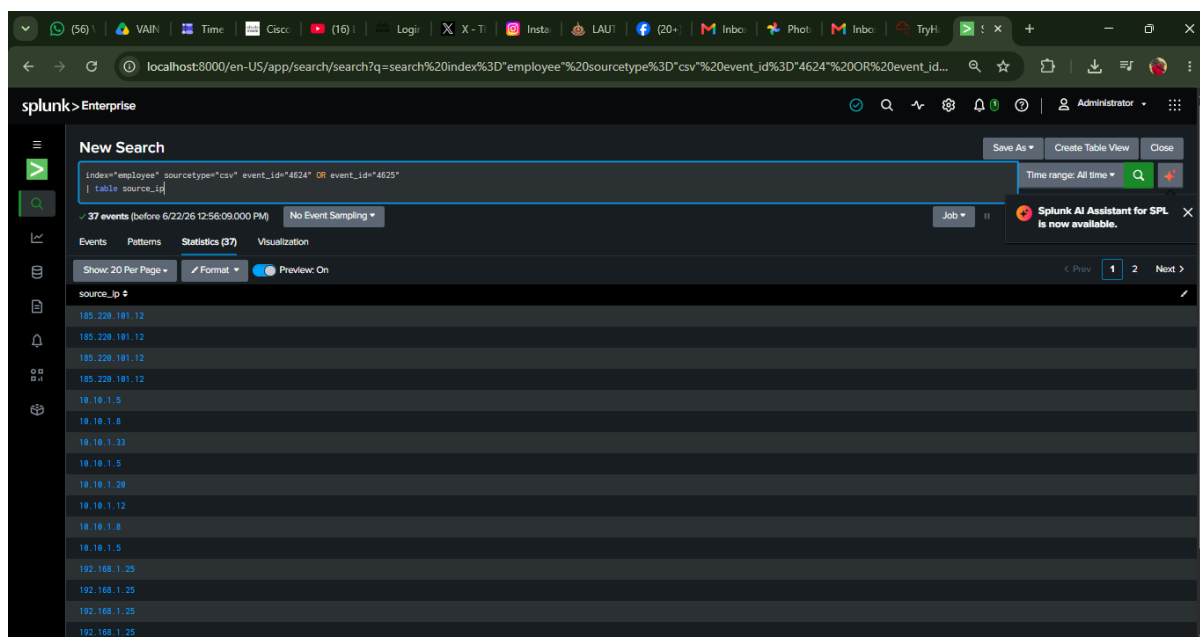
Screenshot 1 – Initial Splunk Alert Investigation

The screenshot shows the Splunk Enterprise interface. The search bar contains the query: `source="C:\Users\OWNER\Downloads\timestamp_log_source.event_id.event (3).csv" host="Employee" index="employee" sourcetype="csv"`. The search results show 157 events. The first event is from 5/12/26 7:35:00 AM, with the message: "2026-05-12 07:35:00,IDS,,,IDS-01,192.168.1.25,45.77.22.91,,,TCP,,,,,ALERT: Threat intelligence match - 45.77.22.91 listed on abuse.ch blacklist". The second event is from 5/12/26 7:05:00 AM, with the message: "2026-05-12 07:05:00,Windows-Security,4624,Successful Login,WS-FIN-04,192.168.1.25,,49980,3389,TCP,svc_backup2,,,SUCCESS,,,Backdoor account still active 6hrs after attack - not yet detected".

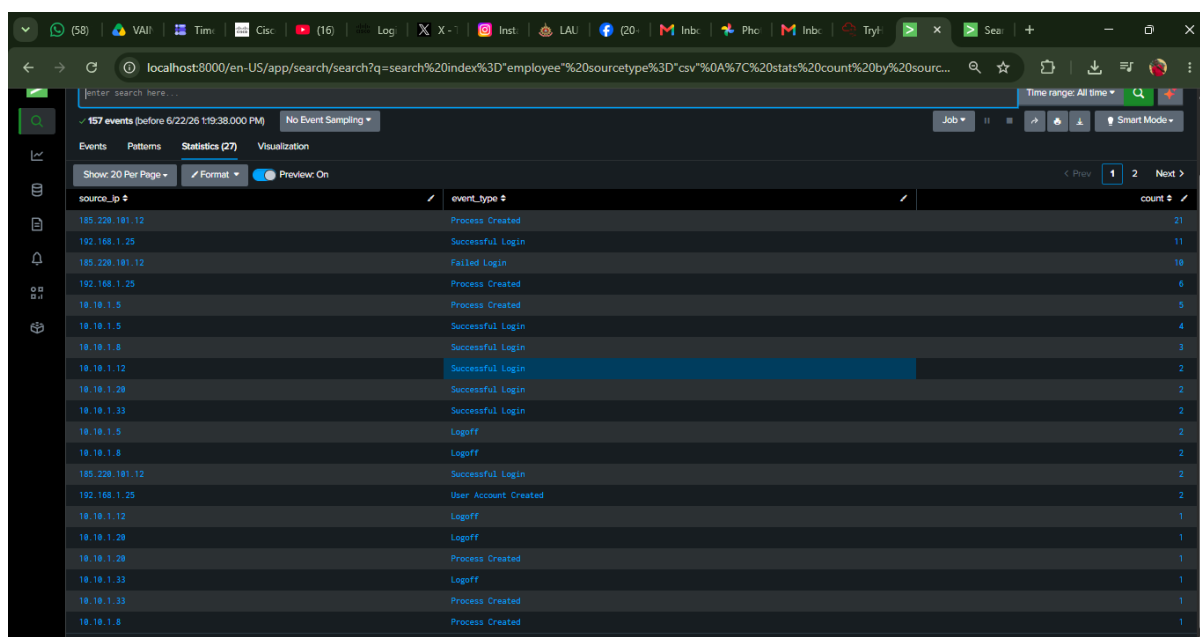
Screenshot 2 – Source IP Identification

The screenshot shows the Splunk Enterprise interface. The search bar contains the query: `index="employee" sourcetype="csv"`. The search results show 157 events. The first event is from 5/12/26 7:35:00 AM, with the message: "2026-05-12 07:35:00,IDS,,,IDS-01,192.168.1.25,45.77.22.91,,,TCP,,,,,ALERT: Threat intelligence match - 45.77.22.91 listed on abuse.ch blacklist". The second event is from 5/12/26 7:05:00 AM, with the message: "2026-05-12 07:05:00,Windows-Security,4624,Successful Login,WS-FIN-04,192.168.1.25,,49980,3389,TCP,svc_backup2,,,SUCCESS,,,Backdoor account still active 6hrs after attack - not yet detected".

Screenshot 3 – Event ID Analysis



Screenshot 4 – Suspicious IP Timeline Review



Screenshot 5 – Authentication Event Correlation

localhost:8000/en-US/app/search/search?q=search%20index%3D*employee%20sourcetype%3D*csv%20source_ip%3D*185.220.101.12%0A%7C...

55 events (before 6/22/26 12:02:00 PM) No Event Sampling

Events Patterns Statistics (55) Visualization

Show: 20 Per Page Format Preview: On

time	source_ip	event_type
2026-05-12 01:01:00	185.220.101.12	
2026-05-12 01:01:05	185.220.101.12	
2026-05-12 01:01:10	185.220.101.12	
2026-05-12 01:01:18	185.220.101.12	
2026-05-12 01:01:25	185.220.101.12	
2026-05-12 01:02:00	185.220.101.12	
2026-05-12 01:02:14	185.220.101.12	
2026-05-12 01:02:22	185.220.101.12	
2026-05-12 01:02:31	185.220.101.12	
2026-05-12 01:02:40	185.220.101.12	
2026-05-12 01:02:49	185.220.101.12	
2026-05-12 01:03:00	185.220.101.12	
2026-05-12 01:13:55	185.220.101.12	
2026-05-12 01:14:22	185.220.101.12	Failed Login
2026-05-12 01:14:35	185.220.101.12	Failed Login
2026-05-12 01:14:48	185.220.101.12	Failed Login
2026-05-12 01:15:01	185.220.101.12	Failed Login
2026-05-12 01:15:14	185.220.101.12	Failed Login
2026-05-12 01:15:27	185.220.101.12	Failed Login
2026-05-12 01:15:40	185.220.101.12	Failed Login

Screenshot 6 – Statistical Analysis and Event Counts

```

kali@kali: ~
Session Actions Edit View Help
Server: 8.8.8.8
Address: 8.8.8.8#53

Non-authoritative answer:
Name: netacad.com
Address: 34.200.167.129
Name: netacad.com
Address: 44.193.149.233
netacad.com nameserver = ns-1911.awsdns-46.co.uk.
netacad.com nameserver = ns-1476.awsdns-56.org.
netacad.com nameserver = ns-240.awsdns-30.com.
netacad.com nameserver = ns-748.awsdns-29.net.
netacad.com
origin = ns-1476.awsdns-56.org
mail addr = awsdns-hostmaster.amazon.com
serial = 1
refresh = 7200
retry = 900
expire = 1209600
minimum = 86400
netacad.com mail exchanger = 30 aspmx3.googlemail.com.
netacad.com mail exchanger = 30 aspmx2.googlemail.com.
netacad.com mail exchanger = 20 alt2.aspmx.l.google.com.
netacad.com mail exchanger = 10 aspmx.l.google.com.
netacad.com mail exchanger = 20 alt1.aspmx.l.google.com.
netacad.com text = "google-site-verification=g7CVgKXjcGaA02xXiZPkst9HPpA9_LY0_Uab0_DRTgc"
netacad.com text = "linkedin-site-verification=f83d41fa-4926-4fc1-bc86-399f34d2ec82"
netacad.com text = "5c9ty312qz1q7yvvly7mmk11nrfpp6kn"
netacad.com text = "v=spf1 include:_spf.google.com include:amazonses.com ~all"
netacad.com text = "identrust_validate=GHH1LQD22HMMen8L8Vz+96QqXOWYASy77U58KT1JnGv"
netacad.com text = "facebook-domain-verification=9a8xflw21o4qxwm9c3pk3d0ete8bu"
netacad.com text = "google-site-verification=TxuIwIjruI4G9oKael5K87LVxJIRJg2v0iy8Rky02Ak"
netacad.com text = "93hd7nffv5d7h3vbwrc14q6n5cjkb2"

```

Investigation Findings

Suspicious IP Address Identified: 185.220.101.12. Multiple failed login attempts, process creation events, and authentication activities were investigated in Splunk. Event correlation and timeline analysis were used to assess potential compromise indicators and support SOC investigation procedures.